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Heat Transfer #3

Date Due: 2/24/26

Acknowledgment: N/A

Date Submitted: 2/22/26

$$1) \dot{E}_{in} + \dot{E}_g - \dot{E}_{out2} - \dot{E}_{out1} = \dot{E}_{st}$$

$$\begin{aligned} \dot{E}_{in} &= \dot{q}_x + \dot{q}_r \\ &= \dot{q}_x \\ &= -kA \frac{\partial T}{\partial x} \\ \dot{E}_{in} &= -k(\pi R^2) \frac{\partial T}{\partial x} \end{aligned}$$

$$\dot{E}_g = \dot{q} A dx$$

$$\dot{E}_g = \dot{q} (\pi R^2) dx$$

$$\begin{aligned} \dot{E}_{out2} &= \dot{q}_r + d\dot{r} \\ &= hP(T_s - T_\infty) dx \end{aligned}$$

$$\dot{E}_{out2} = h(2\pi R)(T_s - T_\infty) dx$$

$$\begin{aligned} \dot{E}_{out1} &= \dot{q}_x + d\dot{q}_x \\ &= -kA \frac{\partial T}{\partial x} + \frac{\partial}{\partial x} (-kA \frac{\partial T}{\partial x}) dx \\ &= -kA \frac{\partial T}{\partial x} - kA \frac{\partial^2 T}{\partial x^2} dx \end{aligned}$$

$$\dot{E}_{out1} = -k(\pi R^2) \frac{\partial T}{\partial x} - k(\pi R^2) \frac{\partial^2 T}{\partial x^2} dx$$

$$\dot{E}_{out1} = -k\pi R^2 \left[\frac{\partial T}{\partial x} - \left(\frac{\partial^2 T}{\partial x^2} \right) dx \right]$$

$$\begin{aligned} \dot{E}_{st} &= \rho c_p A dx \left(\frac{\partial T}{\partial t} \right) \\ \dot{E}_{st} &= \rho c_p (\pi R^2) dx \left(\frac{\partial T}{\partial t} \right) \end{aligned}$$

$$b) -k\pi R^2 \frac{\partial T}{\partial x} + \dot{q} (\pi R^2) dx - h(2\pi R)(T_s - T_\infty) dx + k\pi R^2 \frac{\partial T}{\partial x} - k\pi R^2 \frac{\partial^2 T}{\partial x^2} dx = \rho c_p \pi R^2 dx \left(\frac{\partial T}{\partial t} \right)$$

$$\dot{q} dx - 2h/R (T_s - T_\infty) dx + k \frac{\partial^2 T}{\partial x^2} dx = \rho c_p dx \left(\frac{\partial T}{\partial t} \right)$$

$$\dot{q} - 2h/R (T_s - T_\infty) + k \frac{\partial^2 T}{\partial x^2} = \rho c_p \left(\frac{\partial T}{\partial t} \right) / k$$

$$\dot{q}/k - (2h/Rk) (T_s - T_\infty) + \frac{\partial^2 T}{\partial x^2} - (\rho c_p/k) \left(\frac{\partial T}{\partial t} \right)$$

$$c) \text{ IC: } T(x, 0) = T_i$$

$$\text{ BC: } T(0, t) = T_i$$

$$-k \left(\frac{\partial T}{\partial x} \right) \bigg|_{x=L} = h(T(h, t) - T_\infty)$$

$$2) \dot{E}_m = \dot{E}_{out}$$

$$-k_B A \frac{\partial T}{\partial x} = hA(T_s - T_\infty)$$

$$-k_B (T_s - T_i) \left(\frac{1}{L_B} \right) = h(T_s - T_\infty)$$

$$-30 (T_s - 373.15K) (1/0.1) = 500 (T_s - 293.15K) \rightarrow T_s = 323.15K$$

$$b) \dot{E}_g = \dot{E}_{out}$$

$$\dot{q} A L_1 = -k_B A (T_s - T_i) \left(\frac{1}{L_B} \right)$$

$$\dot{q} L_1 = -k_B (T_s - T_i) \left(\frac{1}{L_B} \right)$$

$$\dot{q} (0.3) = (-30) (323.15 - 373.15) (1/0.1) \rightarrow \dot{q} = 50000 \text{ W/m}^3$$

$$c) -\dot{q}/k_A = \frac{\partial^2 T}{\partial x^2}$$

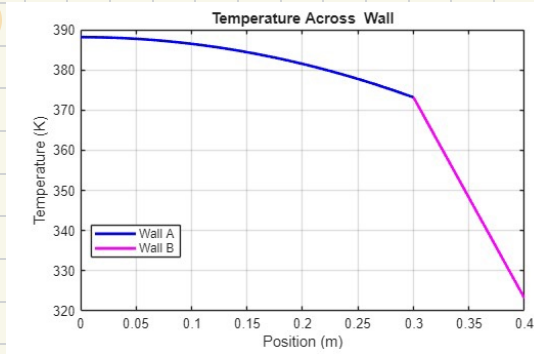
$$-\dot{q} x^2 \left(\frac{1}{2k_A} \right) + C = T$$

$$0 + C = T_w = 328.15K$$

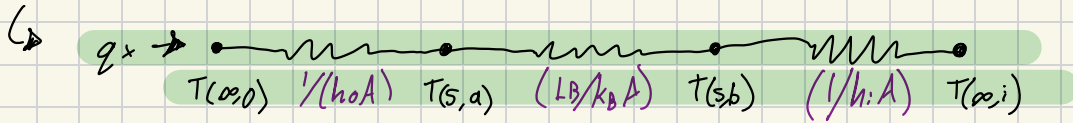
$$T = -\dot{q} x^2 \left(\frac{1}{2k_A} \right) + 328.15K$$

$$373.15 = -50000 \left(\frac{1}{2 \cdot k_A} \right) (0.3)^2 + 328.15 \rightarrow k_A = 150 \text{ W/mK}$$

d.)



3.) $0 \rightarrow a \rightarrow b \rightarrow i$

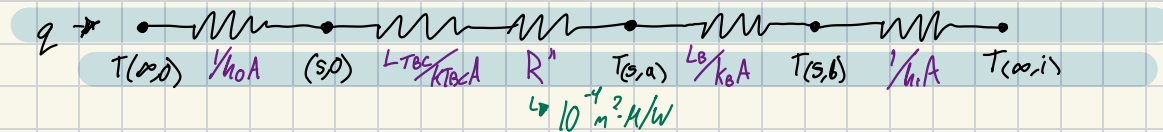


$$R_{tot} = \frac{1}{h_o A} + \frac{L_B}{k_B A} + \frac{1}{h_i A} = \frac{1}{A} (0.0032)$$

$$\frac{(T_{(s, o)} - T_{(\infty, i)})}{(1616 - 4000)} \left(\frac{1}{0.0032 A} \right) = \frac{(T_{(\infty, o)} - T_{(s, a)})}{(1616 - T_{(s, a)})} \left(\frac{1}{0.001 A} \right)$$

$$\Rightarrow T_{(s, a)} = 1236 \text{ K}, \therefore \text{No b/c } 1236 > 1200$$

b.)



$$\frac{[T_{(s, o)} - T_{(s, a)}]}{\frac{1}{A} \left(\frac{L_{TBC}}{k_{TBC}} + \frac{1}{h_o} + R'' \right)} = \frac{[T_{(\infty, o)} - T_{(\infty, i)}]}{\frac{1}{A} \left(\frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R'' + \frac{L_B}{k_B} + \frac{1}{h_i} \right)}$$

$$\frac{[1616 - T_{(s, a)}]}{\left(\frac{0.0005}{2} + \frac{1}{1000} + 10^{-4} \right)} = \frac{[1616 - 400]}{\frac{1}{A} \left(\frac{1}{1000} + \frac{0.0005}{1} + 10^{-4} + \frac{0.005}{25} + \frac{1}{500} \right)}$$

$$T_{(s, a)} = 1104 \text{ K}, \text{ yes b/c } 1104 \text{ K} < 1200 \text{ K}$$